Routes of Administration

- Deep subcutaneous or intramuscular route – most vaccines
- Oral route – Oral BCG vaccine
- Intradermal route – BCG vaccine
- Scarification – Small pox vaccine
- Intranasal route – Live attenuated influenza virus

Types of Immunization

Immunization is of two types:

1. Passive Immunization
2. Active Immunization

1. Passive Immunization

- Passive immunization is produced without challenging the immune system of the body. It is done by administration of serum or gamma globulins from a person who is already immunized to a non- immune person.
- Passive immunization is the administration of preformed antibodies either intravenously or intramuscularly.
- It is used to provide rapid protection in certain infections such as diphtheria or tetanus or in the event of accidental exposure to certain pathogens such as hepatitis B.
- It is also used to provide protection in immune compromised individuals.

Passive natural immunization - acquired from the mother before and after birth. Before birth, immunity is transferred from mother to the fetus in the form of maternal antibodies through placenta. After birth, the antibodies (Ig A) are transferred through breast milk (Table 11.2).

Passive artificial immunization - developed by injecting previously prepared

Table 11.2: Passive Immunization

Infection	Source of Antiserum	Indications
Tetanus	Immune human; horse	Post exposure (plus vaccine)
Diptheria	Horse	Post-exposure
Gas gangrene	Horse	Post-exposure
Botulism	Horse	Post-exposure
Varicella-Zoster	Immune human	Post-exposure in immunodeficiency
Rabies	Immune human	Post exposure (plus vaccine)
Hepatitis B	Immune human	Post-exposure prophylaxis
Hepatitis A	Pooled human Ig	Prophylaxis
Measles	Immune human	Prophylaxis
Snakebite	Horse	Post-bite
Some autoimmune disease	Pooled human ig	Acute thrombocytopenia and neutropenia

TN_GOVT_XII_Micro_Biology_CH11.indd 243 04-03-2019 12:32:04